

CENG381 Stochastic Processes
Final Exam

Duration: 100 minutes

Q1 (25 points). In Gambler's Ruin (A gambler betting on the outcome of a sequence of independent fair coin tosses. If the coin comes up heads, she adds one dollar to her purse; if the coin lands tails up, she loses one dollar. If she ever reaches a fortune of N dollars, she will stop playing. If her purse is ever empty, then she must stop betting.), if $N=4$ and $p=1/2$ (symmetric random walk),

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad P^{30} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{3}{4} & 0 & 0 & 0 & \frac{1}{4} \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} \\ \frac{1}{4} & 0 & 0 & 0 & \frac{3}{4} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

What does this imply?

Q2 (25 points). Let T_{TT} be the (random) number of times we need to flip a coin before we have gotten Tails on two consecutive tosses. Compute the expected value of T_{TT} .

Q3 (25 points). Three frogs are playing near a pond. When they are in the sun they get too hot and jump in the lake at rate 1. When they are in the lake they get too cold and jump onto the land at rate 2. Let X_t be the number of frogs in the sun at time t .

- (a) Find the stationary distribution for X_t .
- (b) Check the answer to (a) by noting that the three frogs are independent two-state Markov chains.

Q4 (25 points). Suppose you arrive at a bank that has two tellers, and suppose that both tellers are busy when you arrive, but there is no-one else in queue. Somebody else arrives right after you, while you are waiting. If service times for teller i are exponential with rate λ_i , $i = 1, 2$, find the expected time you will spend at the bank, under each of the following strategies:

- (a) You pick a teller at random and wait until the customer being served by that teller leaves.
- (b) You will queue up in front of both tellers, and join the first teller that gets freed up.
- (c) What's the expected service time?